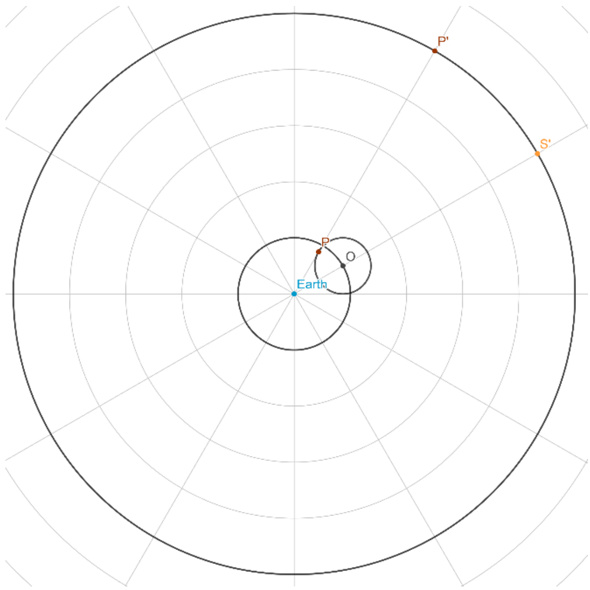
Assignment 3: Chapters 12-17

Due 5pm, 17 April 2026

Answer each of the following questions along with a brief explanation (1-3 sentences) of why it is correct.

1. In Ptolemy's models of planetary motion, what feature(s) of a planets' motion may be captured by an epicycle? Why is there need for further complications such as equant and eccentric?

2. In the figure, a simple epicycle model for a hypothetical planet's motion in a geocentric perspective is shown. The epicycle's center is labelled O. The point O will circle the Earth anti-clockwise with a period, T_O. The planet is labelled P. It will circle the epicycle anticlockwise with a period, T_P. Looking past the planets to the background of stars, the position of the Planet and the Sun along the ecliptic are respectively labelled P' and S'.



   a. Assuming the figure is drawn to scale, what is the elongation of this planet?
   b. Could this planet be a superior planet? If so, what is the value of T_P? If not, why not?
   c. Could this planet be an inferior planet? If so, what is the value, T_O? If not, why not?
   d. Determine the ratio of the epicycle's radius to the deferent's radius, $R_{\mathrm{ep}}/R_{\mathrm{def}}$, based on the planet's elongation.

3. In this question, we make some comparison between Ptolemy's and Copernicus' models for planetary motion.
   a. In Ptolemy's System, the planets are (mostly) ordered according to their tropical periods. Is this supported by observations or by philosophy?
   b. In Copernicus' System, the planets can be ordered according to their sidereal periods. Is this supported by observations or by philosophy?

4. The benefits of Copernicus' model arise from the hypothesis of the moving Earth. Today, the moving Earth hypothesis enjoys very broad acceptance.
    a. What evidence did Copernicus have to support the moving Earth hypothesis?
    b. Based on your answer to part a., was Copernicus' model more scientifically supported than Ptolemy's? (HINT: One good way to say something is scientifically supported is to show that it is supported by practice of the scientific method)